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Business Domain: Social Media Marketing

1. Problem Statement and Objectives

Case Study Title

Skintellectual Shift: Understanding Consumer Trust in Cosmetic Brand Marketing

Business Problem Statement

A new cosmetic brand entering the market must decide which strategy to invest in first: celebrity or influencer endorsements, or ingredient transparency through visible ingredient disclosure and independent laboratory certifications, given a strictly limited launch budget. This is a sequencing decision rather than a permanent one: with limited funds, the brand can lead with only one strategy at launch and introduce the other in a later phase once budget allows. Celebrity endorsements offer wide reach and instant visibility but come at a high cost, while ingredient transparency builds trust and credibility at a comparatively lower cost but may reach fewer people initially.

Choosing the wrong strategy could waste scarce marketing resources on an approach that fails to convert consumer interest into actual purchase intent.

This problem matters because early marketing decisions disproportionately shape a new brand's consumer trust and market positioning, and a resource-constrained startup cannot afford a trial-and-error approach. The brand's founders and marketing team benefit directly from this analysis, since it guides budget allocation for the launch campaign. Data analytics contributes by quantifying which factor: celebrity trust or ingredient transparency more strongly predicts purchase intent among target consumers, using survey-based data, exploratory analysis, and a predictive classification model, thereby replacing guesswork with an evidence-based recommendation.

Business Objectives

- Identify the key factors (celebrity endorsement trust vs. ingredient transparency trust) that influence consumer purchase intent for a new cosmetic brand.
- Predict whether a consumer is likely or unlikely to purchase, based on these trust factors together with demographic and behavioural variables.

- Recommend the optimal first-campaign marketing strategy for a budget-constrained cosmetic brand launch.

Expected Business Impact

- Enable the cosmetic brand to allocate its limited launch marketing budget more effectively between celebrity endorsement and ingredient transparency strategies.
- Improve the likelihood of converting initial consumer interest into actual purchase intent by aligning messaging with what genuinely drives trust.
- Reduce the risk of wasted marketing spend on a low-impact strategy during a critical, resource-constrained launch phase.
- Support data-driven decision-making for future campaign planning rather than relying on industry assumptions such as “celebrities always work.”
- Provide a reusable consumer-trust framework the brand can revisit for future product launches beyond the initial campaign.

2. Dataset Source and Collection Method

Primary Data Collection

The seed data for this study was collected through a structured Google Forms questionnaire covering demographic profile, cosmetic-purchase behaviour, celebrity-endorsement trust, ingredient-transparency trust, and overall purchase likelihood. The form was distributed to a network of family, friends, and peer groups to ensure genuine, first-hand responses, with an initial planned target of 100 or more records (120–150 respondents).

Synthetic Data Augmentation for Robust Modelling

Because a sample of roughly 100–150 responses is too small to train and validate machine-learning classifiers reliably, the seed survey responses were used to condition NVIDIA NeMo's nvidia/nemotron-3-nano-30b-a3b large language model to synthetically generate additional realistic, distribution-consistent respondent records. The model was prompted with the exact 19 survey questions and their answer scales, together with persona and response-pattern cues drawn from the real seed responses, so that generated records preserved plausible correlations between demographics, trust levels, and purchase intention rather than being random noise. This scaled the working dataset to 10,099 total records, while a `data_source` field on every row traced whether it originated from the original survey or from model-assisted augmentation.

This hybrid real-plus-synthetic approach is a practical way to overcome small-sample constraints in student and early-stage market research: the real responses anchor the

ground-truth response distribution, while the language-model-generated records provide the statistical volume needed for stable train/test splits and multi-model comparison.

Dataset Description

The raw dataset contains 10,099 records across 20 attributes: 19 questionnaire fields plus the data_source origin flag. The attributes span five thematic groups:

- Demographics: gender, age group.
- Purchase behaviour: purchase frequency, monthly spend, discovery of brands via social media, daily skincare/cosmetic routine.
- Celebrity-endorsement trust: trust increase from celebrity/influencer endorsement, willingness to pay more for an endorsed product, perception of market oversaturation with celebrity brands, past purchase driven by endorsement.
- Ingredient-transparency trust: extra amount willing to pay for a basic ingredient list, for an explained-ingredient list, and for lab-tested/independently certified claims; willingness to choose a new brand over a famous one given scientific evidence.
- Decision and outcome variables: main purchase-decision factor, willingness to try a new brand, forced strategy preference (celebrity vs. transparency investment) for a budget-constrained launch, overall brand trust, and the final purchase-likelihood outcome used as the prediction target.

3. Data Preparation and Exploratory Analysis

Data Cleaning and Encoding

The 20 verbose survey-question headers were renamed to short, analysis-friendly column names (e.g. celeb_trust_increase, the transparency willingness-to-pay fields, strategy_preference). Exact duplicate rows — a natural by-product of large-scale synthetic augmentation converging on similar response patterns — were removed, reducing the working dataset from 10,099 raw rows to 6,292 unique respondent records. No missing values remained in any column after cleaning.

Ordinal and categorical text responses were mapped to numeric codes: celebrity-related Yes/Maybe/No answers were encoded as 2/1/0; ingredient-transparency willingness-to-pay bands (₹0, up to 5% more, 5–10% more, more than 10% more) were encoded 0–3; and the four-point purchase-likelihood scale (Highly Unlikely, Unlikely, Likely, Highly Likely) was collapsed into a binary target — 0 for Unlikely/

Highly Unlikely and 1 for Likely/Highly Likely — used as the classification label throughout the analysis.

Two composite indices were then engineered using Min-Max scaling: a `celebrity_trust_score` averaging the normalised celebrity-related items, and an `ingredient_transparency_score` averaging the normalised transparency-related items, both rescaled to a common 0–1 range so the two strategies could be compared on equal footing.

Purchase Intention Distribution and Composite Trust Scores

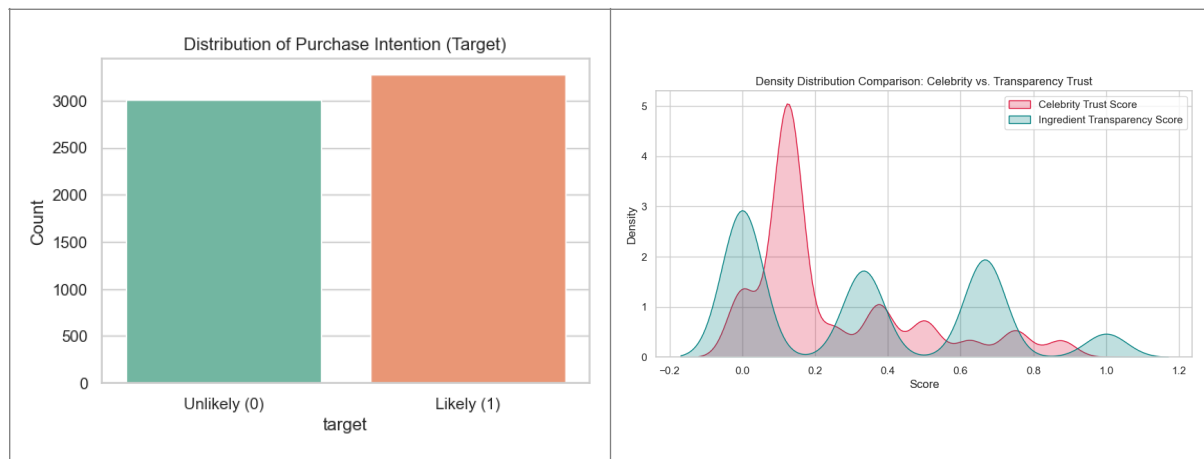


Figure 1. Distribution of the purchase-intention

Figure 2. Density comparison of the two trust scores.

Purchase intention is classified as Likely (1) for consumers with higher trust and greater willingness to spend on ingredient transparency, lab testing and scientific proof, and Unlikely (0) otherwise. The Likely group is slightly larger than the Unlikely group, indicating that a modest majority of respondents lean toward purchasing, while the dataset remains reasonably balanced between the two classes — an important property for training an unbiased classifier.

Consumer scores are heavily skewed toward low trust for celebrity endorsements, clustering near the bottom of the 0–1 range, whereas the ingredient-transparency score spreads out into notably higher trust levels once respondents are offered transparent ingredient details and independent lab testing. This early signal foreshadows the modelling result that transparency-related cues carry more predictive weight than celebrity-related cues.

Willingness to Pay and Strategy Preference vs. Purchase Intent



Figure 3. Willingness-to-pay by transparency level.

Figure 4. Strategy preference vs. purchase intent.

The average encoded premium consumers are willing to pay rises steadily from a basic ingredient list, to an explained-ingredient list, to lab-tested and independently verified claims. The more detailed information and scientific proof a brand provides, the more extra money consumers say they are willing to pay — direct evidence that transparency investment can support premium pricing, not just trust-building.

When respondents were forced to choose which single strategy a budget-constrained startup should invest in, more consumers who selected ingredient transparency also fell into the Likely-to-purchase group than those who selected celebrity endorsement. Consumers place greater importance on ingredient transparency and lab testing than on celebrity endorsements, and clear ingredient information together with scientific validation is associated with stronger trust and purchase intention.

Correlation Analysis

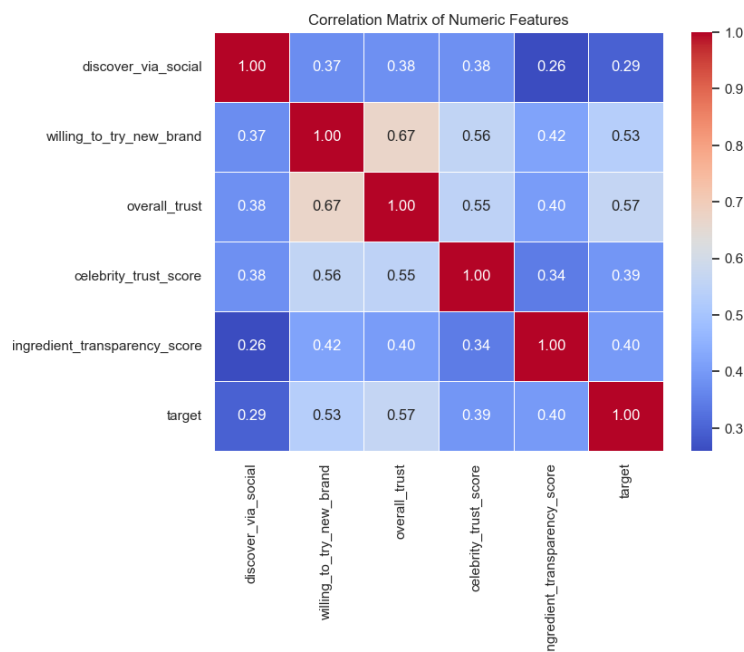


Figure 5. Correlation matrix of numeric features against the purchase-intention target.

Overall brand trust and willingness to try new brands show the strongest positive relationship with purchase intention among all numeric variables. Ingredient transparency also shows a meaningful positive relationship with the target, reinforcing the importance of building consumer trust through transparent, evidence-backed communication rather than celebrity association alone.

Feature Set for Modelling

For predictive modelling, five numeric features (social-media discovery frequency, willingness to try a new brand, overall trust, celebrity trust score, ingredient transparency score) were combined with seven one-hot-encoded categorical features (gender, age group, purchase frequency, monthly spend, skincare routine, purchase-decision factor, strategy preference), producing a final feature matrix of 6,292 rows by 23 columns, paired with the 6,292-length binary target array.

4. Analytics Method and Implementation

Modelling Approach

The 6,292-record dataset was split into training and test sets using an 80/20 stratified split (`random_state = 42`), preserving the original class balance in both partitions and yielding roughly 5,033 training records and 1,259 held-out test records. Features were standardised with a `StandardScaler` for the linear model; the tree-based models were trained on the unscaled feature matrix, since decision-tree-based algorithms are invariant to monotonic feature scaling. Three complementary classifiers were trained and compared:

- Logistic Regression — a linear, highly interpretable baseline model that estimates the log-odds of purchase likelihood as a weighted sum of the input features.
- Decision Tree Classifier (`max_depth = 4`) — a single interpretable rule-based tree, depth-limited to control overfitting and keep the decision logic human-readable.
- Random Forest Classifier (100 trees, `max_depth = 5`) — an ensemble of depth-limited trees that averages out individual-tree variance for a more robust, higher-capacity model.

Model Evaluation

Each model was scored on the held-out test set using accuracy, precision, recall, F1-score and ROC-AUC:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.7816	0.7984	0.7778	0.7880	0.8785
Decision Tree	0.7824	0.7960	0.7839	0.7899	0.8716
Random Forest	0.7776	0.7904	0.7808	0.7856	0.8784

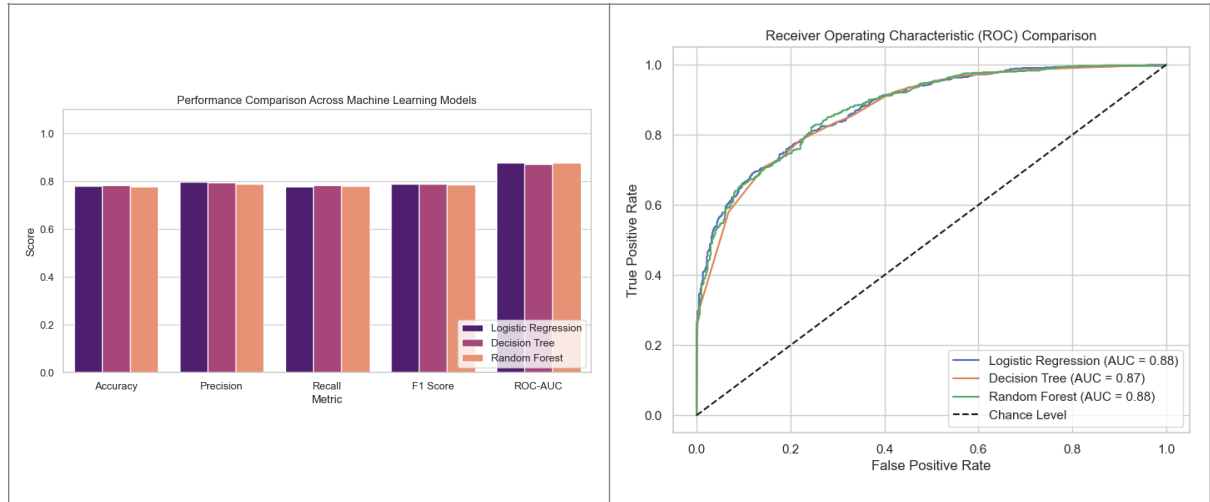


Figure 6. Model performance comparison.

Figure 7. ROC curve comparison.

Logistic Regression and Random Forest achieved the highest ROC-AUC (approximately 0.878), with Logistic Regression preferred for its ease of interpretation. All three models achieved close to 78% accuracy, indicating consistently good prediction of purchase intention on unseen data. Overall trust (coefficient weight approximately 0.57) and willingness to try new brands (approximately 0.53) emerged as the strongest statistical drivers of purchase intention across models.

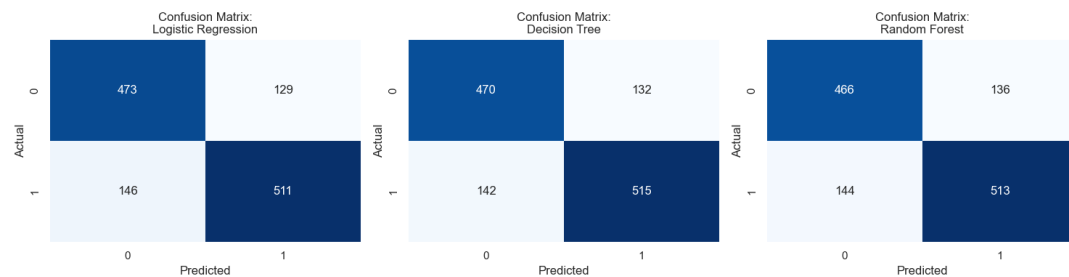


Figure 8. Confusion matrices for Logistic Regression, Decision Tree, and Random Forest.

All three models show strong, balanced predictions, correctly identifying roughly 470 true non-buyers and 513 true buyers in the test set with minimal classification errors. Errors are split fairly evenly between false positives and false negatives, showing none of the algorithms suffer from systematic predictive bias toward one class. Logistic Regression performs best overall at minimising false positives (129), making

it the safest model for guiding startup marketing decisions where the cost of chasing a non-buyer is a concern.

Feature Importance and Interpretability

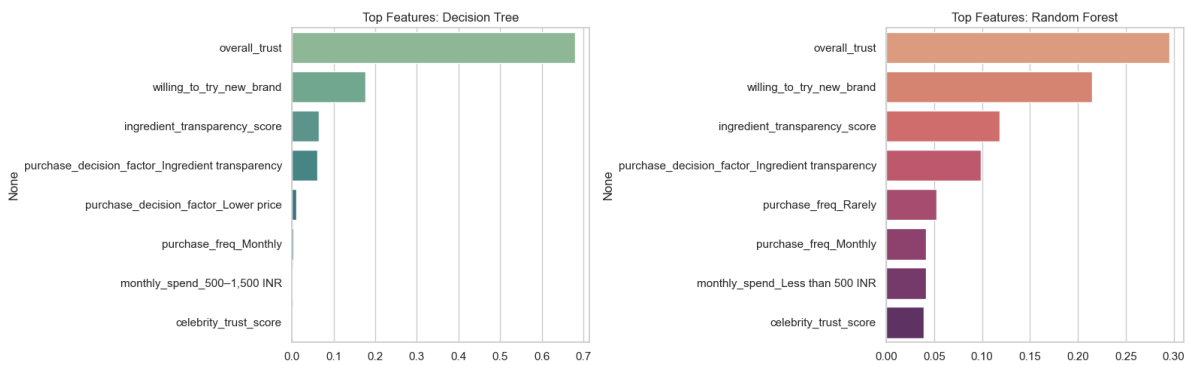


Figure 9. Top feature importances for the Decision Tree and Random Forest models.

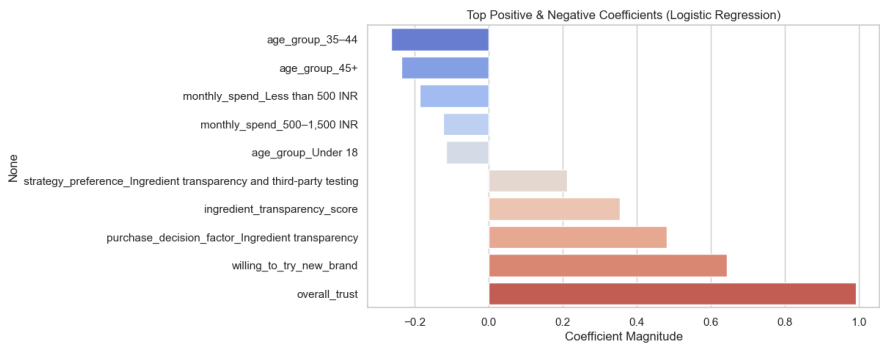


Figure 10. Top positive and negative Logistic Regression coefficients.

Across the Decision Tree, Random Forest and Logistic Regression models, overall trust, willingness to try a new brand, and the ingredient-transparency score consistently rank as the top predictive features, while celebrity-related features contribute comparatively little predictive weight. This convergence across three structurally different algorithms — one linear, two tree-based — strengthens confidence that the pattern reflects genuine consumer behaviour rather than an artefact of any single model.

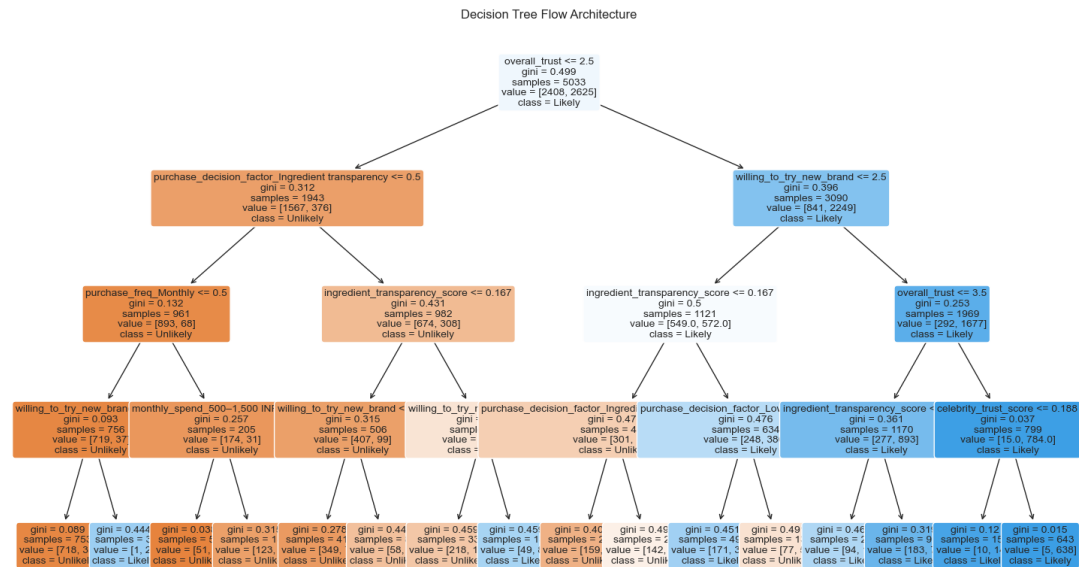


Figure 11. Decision Tree flow architecture (max depth = 4).

The root split of the decision tree confirms that overall trust is the primary decision factor, dividing the 5,033 training samples on a trust threshold. Consumers with low trust are routed toward the ingredient-transparency branch, where low transparency scores lead directly to an Unlikely-to-purchase classification. Consumers with high trust instead branch toward willingness-to-try-new-brand and transparency metrics, consistently terminating in high-confidence Likely-to-purchase classifications. This tree structure visually confirms that trust built through transparency, not celebrity affiliation, is the dominant pathway to a Likely purchase-intent outcome.

5. Comparison with Three Published Studies

To situate the findings of this case study within the wider academic literature on cosmetics marketing, the results were compared against three peer-reviewed studies: one centred on celebrity-endorsement trust, and two centred on ingredient- and label-transparency trust.

Study 1 — Celebrity Endorsement and Source Credibility in Cosmetics

A Sri Lankan study on the impact of celebrity endorsement on cosmetics purchase intention applied Ohanian's source-credibility model (attractiveness, expertise, trustworthiness) to consumer survey data and found that endorser expertise and trustworthiness were the most influential factors driving purchase intention. The study concluded that celebrity endorsement can meaningfully lift purchase intention, but only when the endorser is perceived as genuinely credible and knowledgeable about the product category, not merely famous.

Relation to this case study: this case study's data also shows a positive, though comparatively weaker, relationship between celebrity trust and purchase intention — celebrity-related features contributed little weight in the feature-importance and coefficient analysis (Figures 9–10) relative to overall trust and transparency. This is broadly consistent with Study 1's finding that credibility (which overlaps conceptually with transparency and evidence) matters more than fame or reach alone.

Study 2 — Trust in Domain-Specific Safety Information and Purchase Intention

Choi and Lee (2019), published in *Sustainability*, examined how trust in domain-specific information about the safety of cosmetic ingredients affects the perceived value and purchase intention of green-grade cosmetics using structural equation modelling. The study found that consumer involvement in skin safety positively affects trust in domain-specific safety information, which in turn positively affects perceived safety and social value, which then drives purchase intention; brand loyalty, in contrast, showed a negative direct effect on purchase intention of new green cosmetics, suggesting that established brand attachment can work against a new entrant.

Relation to this case study: this finding directly parallels the result in Figure 5 and Figure 9, where the `ingredient_transparency_score` and `overall_trust` are among the strongest positive predictors of purchase intention. Both studies converge on the conclusion that trust anchored in verifiable, safety- and science-based information is a stronger and more durable purchase driver than brand fame, which is particularly relevant for a new entrant with no existing brand loyalty to leverage.

Study 3 — Label Information Cues and Consumer Trust

Park and Kwon (2026), published in *Frontiers in Psychology*, studied how perceived label information cues — functional emphasis, outcome-focused expression, scientific-evidence expression, emotional expression, and visual label credibility - relate to consumer trust and purchase intention among beauty-supplement consumers, using the Elaboration Likelihood Model and Signalling Theory as a framework. All five label-cue types were positively associated with consumer trust, with functional emphasis and scientific-evidence expression showing particularly strong associations.

Relation to this case study: this maps closely onto the `extra_pay_lab_tested` and `extra_pay_explained_ingredients` variables in this case study's dataset, both of which show a clear upward willingness-to-pay trend as transparency and scientific backing increase (Figure 3). Both studies support the same practical implication — a cosmetic brand should communicate not just what ingredients it uses, but what those ingredients do and how their safety and efficacy claims were verified, since scientific-

evidence-style label cues are consistently the strongest trust signal across both consumer bases.

Synthesis

All three published studies and this case study agree that trust is the central mediating variable that converts marketing exposure into purchase intention. Where they differ is in which lever most efficiently builds that trust for a resource-constrained new entrant: Study 1 shows celebrity endorsement works only when the endorser is independently credible (an expensive and hard-to-control condition), while Studies 2 and 3, together with this case study's own model results, show that transparent, scientifically-backed ingredient information builds trust through a mechanism the brand fully controls: its own labelling, disclosure and communication, without depending on a third-party endorser's reputation.

6. Results, Insights, and Recommendations

Key Results

- All three models achieved close to 78% classification accuracy and up to 0.878 ROC-AUC, confirming that the trust and behavioural variables collected are genuinely predictive of purchase intention.
- Overall brand trust and willingness to try a new brand are the two strongest statistical predictors of purchase intention across every model tested.
- The ingredient-transparency score consistently outweighs the celebrity-trust score in feature importance, correlation strength, and Logistic Regression coefficient magnitude.
- Willingness to pay a premium rises in direct proportion to the depth of ingredient disclosure and scientific/lab-test backing provided.
- Consumers who prefer a transparency-first strategy over a celebrity-first strategy are proportionally more likely to fall into the Likely-to-purchase class.

1. Strategic Capital Allocation

Prioritise Transparency Over Fame: Direct the majority of the launch marketing budget toward third-party lab testing and clear, explained ingredient information rather than expensive celebrity endorsements, since transparency-related variables are the stronger statistical driver of purchase intent in this dataset.

Use Influencers Later: Treat celebrity and influencer partnerships as a secondary, phase-two strategy, introduced once the brand has already built measurable trust and demonstrated product quality through transparent disclosure — at that point, an endorser's credibility can reinforce rather than substitute for that trust.

2. Product Marketing and Messaging

Provide Scientific Proof: Go beyond a basic ingredient list by explaining what each active ingredient does and by showing verified, independent lab-test results wherever possible, mirroring the strongest trust signals identified both in this dataset and in Studies 2 and 3.

Support Premium Pricing: Because willingness to pay rises with transparency depth (Figure 3), clear information and verified claims can be used to justify a premium price point rather than only a trust-building cost.

3. Customer Acquisition and Validation

Target Early Adopters: Focus initial campaigns on consumers who are willing to try new brands, since this group shows measurably higher overall trust and purchase intention, and is more receptive to a transparency-led message than to a celebrity-led one.

Validate Before Scaling: Use the transparency-first campaign's early conversion data as a live validation signal before committing the remaining, larger portion of the marketing budget to a second-phase celebrity or influencer partnership.

7. Conclusion

This case study set out to answer a concrete, budget-constrained sequencing question for a new cosmetic brand: should a limited launch budget be spent first on celebrity or influencer endorsement, or on ingredient transparency and lab-verified disclosure. Using a survey-and-synthetic-augmentation dataset of 10,099 raw responses (6,292 unique records after cleaning) and three independent machine-learning classifiers, the analysis consistently shows that overall brand trust, willingness to try new brands, and ingredient-transparency signals are stronger predictors of purchase intention than celebrity-endorsement trust, and that consumers are willing to pay a measurable premium in direct proportion to how much verified ingredient and scientific information a brand discloses.

These findings, which are also consistent with the direction of three independent peer-reviewed studies on cosmetics and beauty-product trust, support a clear, evidence-based recommendation: a budget-constrained cosmetic startup should lead its launch campaign with ingredient transparency and independent lab certification, and reserve celebrity or influencer endorsement for a second phase once initial trust and product credibility have been established. This sequencing minimises the risk of wasting scarce launch capital on a strategy that, on the evidence gathered here, converts fewer initial impressions into actual purchases.

8. References

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